Gamified Smart Mirror to Leverage Autistic Education – Aliza

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***Abstract*—Autism is a neurodevelopmental disorder that causes difficulties in communication, emotional responsiveness and social skills. There has been a global increase rate in autism and lack of resources locally to educate ASD children. As this condition affects children at an early stage, it remains a challenge in learning. Even though today's world there are ample of teaching methods and technologies, people are unaware of the use and impact of them. This paper presents "Aliza" Gamified smart mirror to teach basic education for autism children. "Aliza" consists of four core components such as writing mentor for pre-writing, math tutor for mathematics, verbal trainer for speech and attentiveness tracker for emotion detection. These components assist and enhance their competency in education. The users of the "Aliza" will be constantly monitored and evaluated during their training using Convolutional Neural Network (CNN). The interactive games are given to impact their learning process while the generated report from the Deep Learning evaluation system can acquaint parents and the tutors with the progress of the children. Through this research, it is expected to improve autistic children's basic education with assistance of "Aliza".**

Keywords— Aliza, smart mirror, Convolutional Neural Network (CNN), emotion detection, pre-writing, Autism Spectrum Disorder (ASD), Deep Learning

I. INTRODUCTION

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that is characterized by heterogeneous deficits in social communication, interaction and pattern of behavior across individuals. According to world health organization, one in 160 children has ASD worldwide and its prevalence appears to be increasing globally[1]. Despite the impairments, the individuals with ASD are succeeding in different sectors with the right support and early intervention. Autism is not a learning disability yet it can affect their learning skills. Since Autism children have strong visual processing skills, using pictures and technology can be more effective to improve their learning skills than the traditional methods. Technological advances have potentially lead to novel and more effective treatment strategies and enhance quality of life for people with ASD and their families [2]. Several studies showed that most people with autism show a natural affinity for technology and a good disposition for using technology and learning through the use of computers [3] as the experiences it provides are predictable and structured, which helps people with ASD to maintain their routines and repetitive behavior without affecting their comfort [4].

This paper presents "Aliza", a smart mirror with series of activities that would help to develop different skills of the children with ASD. A research "Engaging with the self" [5] proves that ASD children are more engaged with mirror than the children with Down syndrome during the self-recognition test. The system integrates with three main skill development programs and an attention monitoring functionality. The smart mirror provides pre-writing skills such as drawing shapes with lines and curves. It is also done in a step by method starting by connecting dots to complete shapes, lines, curves and patterns. Gradually then based on the shapes, lines and curves letters are taught to be written. Math skill development program includes activities for counting, number identification as well as number writing. It is well-known that autism children have language and communication challenges. Therefore, the system provides verbal activities to make them speak. They easily tend to get distracted and seem tedious. Therefore, throughout the activities, the child will be monitored and mirror react upon their emotions. These four functionalities with an additional feature of visualizing the report of children's performance would help improving their skills.

II. RELATED WORK

Technology usage has prominently risen in intervention and researches of ASD. People with autism are often overwhelmed by the demands of human interaction. To beat this problem, number of researches have started to explore ways to use AI to teach Autism children [6].

Alvaro Fernandez-Lopez [7] devised a mobile platform based on iPad and iPod touch devices called Picaa for people with special needs. Picaa is a platform to design educational activities such as Exploration, Association, Puzzle and Sorting activities. Its aim is to help special needs children to learn basic education. Otsimo [8] is a mobile application that helps the ASD students to learn numbers, colors and shapes. Autism Read & Write [9] is an app with simple activities like finding given object's name. Even though these applications provide activities and games like Aliza, it fails to provide an integrated solution to develop multiple skills.

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Thierry et al..[10] proposed an Online Handwritten Shape Recognition systems using segmental hidden markov models (HMM). The system can handle two-dimensional graphic shapes such as Latin and Asian characters, command gestures, symbols, thumbnails and geometric shapes. At the end of the proposed system above, they found that the performance achieves 90% accuracy with a simple model with only one mixture component and that as the sample size increases with the three mixture components in the character models, the performance increases accordingly and thus the performance achieves a maximum accuracy of 93%. However, the have some drawbacks such as this proposed system get less accurate for some specific applications and layers are not trained globally therefore the system is not optimal.

Acapela group have invented a robot with facial expressions and a screen to display the instruction. MILO's functionalities are turn-on emotions, join attentions, speaking activities, improve social behaviors and imitations. MILO is more like a peer who will always keep interacting with the ASD child. Only humans will get bored or feel lazy to do things repetitively, however milo is a robot that doesn't have any feelings such as laziness or tired. It will always keep interacting with the student with recurring positive enforcement [11]. The objective of the robot is to develop social and behavior skills by continuously encouraging positivity for the ASD student. This is only an interactive companion, teaching emotions and social communication to Autism children.

The research conducted by Ernst K and Tatiana B named Improved method of handwritten digit recognition tested on MNIST database uses a neural classifier LImited Receptive Area (LIRA) for the image recognition [12]. It contains three main neuron layers sensor, associative and output. The sensor layer is connected with the associative layer with no modifiable random connections and the associative layer is connected with the output layer with trainable connections [12]. This system was tested with MNIST database which contains 60,000 handwritten digit images and a dataset with 441 images of assembly micro device. The test results contributed a least error rate of 0.61%. Overall the LIRA classifier converges sufficiently fast and produces sufficiently good results.

Humanoid robot NAO, created by Softbank Robotics has its capability to blink its eyes, speak and play music which can entice the children's interest to engage in communication [13] .Both humanoid robot Milo and NAO only focus in training social and communication skills.

The research conducted on computer game to enhance speech problems for Autism children[14] provides learning activity which encourages to repeat the spoken word with the system. The voice recognition system in this app uses DTW (Dynamic Time Warping) for Spanish and SPHINX4 for English. The preprocessing of the signal is done using MFCCs (Mel-Frequency Cepstral Coefficients). 23ms size window is used for the fragmentation in the windowing algorithm and a 10ms time displacement. The correctly preprocessed MFCC compares with its equivalent one stored in the database to check which one is close to the obtained word in DTW algorithm. It calculates the time independent similarities between two vectors of the signal. The other technique SPHINX4 is a framework based on statistical recognition using HMM (Hidden Markov Model) and each associating HMM with single sound unit. The other technique SPHINX4 is a framework based on statistical recognition using HMM and each associating HMM with single sound unit.

Moreover, there are wide ranges of applications which are game oriented to develop their different skills. Finkelstein et al..[15] designed a game "CMotion" that will use virtual humans to teach emotion recognition and programming concepts to Autism children. This is designed to teach the intended users how to recognize facial expressions and manipulate an interactive virtual character using a visual drag-and-drop programming interface. They found that the system of virtual human teaching has increased learning capacity and learning interest than a system of regular classroom teaching to Autism children. Ould Mohamed [16] developed an educational personalized games activity with behavior tracker during the game session. It helps to Autism children to focus their attention on a specific task.

Real Time Face Expression Recognition of Children with Autism research conducted by Suzan and Mariofanna, states about a system to recognize facial expressions in children with ASD during playtime[17]. The emotion recognition in this system uses web camera real-time video stream input of the user and a Viola Jones frontal face for the initial facial detection. Then detected face is processed through multi resolution active shape model which tracks 116 facial points of the face. Then a Support Vector Machine (SVM) classifier detects the facial emotion based on the facial points. All 7 basic emotions are tracked by this system and has a 93% of success rate. Even though it predicts expressions accurately it misclassifies fear-surprise and sadness-disgust as those expressions trigger with similar facial points.

We propose a system that provide multiple games for skill development considering the lacking factors in the previous researches and products. Applying Smart mirror concept for autism education is a new approach and could create an impact as per the research study [5].

III. Methodology

A. Data Collection

This study gathered primary data from randomly selected 100 ASD children between the age of 5 to 12 from 5 selected schools in Jaffna districts. The data was collected in the form of document, audio from selected ASD children's routine lessons. These collected data will be kept confidential and secure due to privacy concern of the children. In addition, publicly available data was used to train some of the models.

B. Hardware and Software components of the Aliza

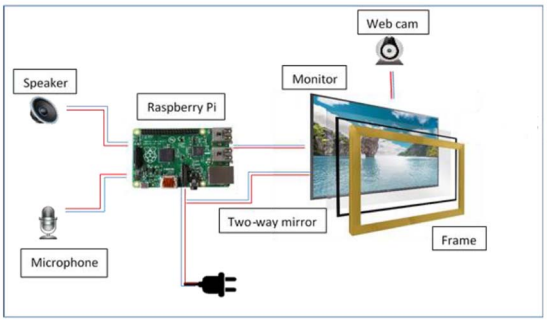


Fig. 1. Hardware Componenets

As shown in Fig. 1, Two-way mirror with 70% reflection on top of LED monitor, covered with a wooden frame to hold them together. The LED monitor is to display the program.

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These two aesthetic components are 20 inches. A speaker and web camera in-built with microphone is connected to the Raspberry Pi for voice recognition and emotion recognition. Raspberry Pi is the brain of smart mirror which has the processing capability with low processing power. It is booted up with Linux 64-bit OS which is configured to run Magic mirror module during the startup. Inside the Magic mirror module, our application resides as a game that could be played by the user.

a) Writing Mentor: Writing skill is used for self-expression, communication, and recording of thoughts and experiences. Informal reports acknowledge that children with autistic spectrum disorder (ASD) have higher levels of writing disability. Also, learning pre-writing skills through game, can further enhance the sense of presence and engage even more deeply with ASD children. Therefore it's consequently, further foster motivation to learn pre-writing skills [18].

In this research, an advanced self-teaching system is introduced with the help of a smart mirror (Aliza). This writing mentor activity games are based on tracing and connecting dots strategy. Through this, ASD children are easily able to develop their pre-writing skills [19].

In the first step before starting, system will pre-evaluate the writing skills of ASD children with the help of CNN model. After the pre-evaluation process, the system will provide some pre-writing skill training activities such as lines, curve and shapes drawing games to ASD children. The system continuously evaluates those children when they are practicing and prepare a progress report.

Pre-writing activity's levels will be levelled up through every evaluation report. For example, after the 1st level of activity completion, the system will give a task to a user based on 1st level activities. If the evaluation report shows the progress level above 80%, then the activity level is levelled-up, if not the same level activities will continue until the user achieve the progress level 80%.

To evaluate user's pre-writing skills and prepare a progress report, a seven-layered Convolutional Neural Network (CNN) with one input layer followed by five hidden layers and one output layer is tried out. It's illustrated in Fig. 2.

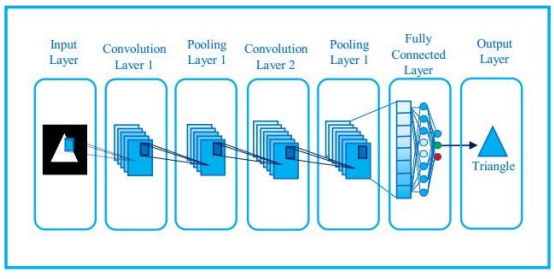


Fig. 2. Seven-layered CNN

The input layer consists of 300 by100 pixel images. The input pixels are grayscale with a value 0 for a white pixel and 1 for a black pixel.

Here, this model of CNN has five hidden layers. The first hidden layer is the convolution layer 1 which is responsible for feature extraction from an input data. The next hidden layer is the pooling layer 1. It reduces the output information from the convolution layer and reduces the number of parameters and computational complexity of the model. Convolution layer 2 and pooling layer 2 which has the same function as convolution layer 1 and pooling layer 1 and operates in the same way except for their feature maps and kernel size varies. A fully connected layer (dense layer) is another hidden layer. It is fully connected and connects every neuron from the previous layer to the next layer. In order to reduce over fitting, dropout regularization method is used at fully connected layer. The output layer of the network classifies the output as the lines, curve, and shapes types. Through this seven-layer CNN strategy can identify the input image, as well as it can produce the output of percentage which is consistent with the ideal image.

b) Basic Math Tutor: Math mentor for Aliza has four games basics, counting, identifying and writing.

Initially math mentor begins from the basic game, its displays a number with text and audio to train them identification of digits. Next the counting game, the user is given objects to identify and count the objects to predict the number. The 3rd game is identification of missing numbers. These games are in a level basis that gear towards difficult levels and are given to prep up the students for the main goal of self-written numbers.

A CNN model is used in the final game, where the canvas is displayed to write a number and then evaluated the precision of the number. The number written in the canvas passes through out 5 convolution neural network where the image is resized and processed to generate the output shown in Fig. 3 given below.

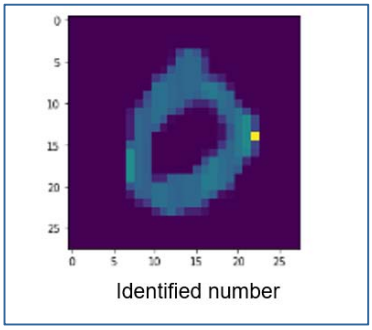


Fig. 3. Output of Math Tutor game's CNN model

Then system flattens and dense the image to train the model more accurately. Each captured picture has its own pixel, considering the edges of the written number every edge will have a pixel weight, each pixel gets assigned to a neuron in the CNN layer initially. In the upcoming layers the weight of the neuron is being multiplied by a constant. At the end, algorithm outputs a probability distribution, according to the percentage that match with the sum of weights the number will be determined. It simplifies the probability distribution using NumPy library into a simple number.

At the end of the process, a report will be auto generated with the progress of the ASD student comparing with a non ASD student. Rather than the traditional teaching method this will be a new dimension of autism education.

c) Verbal Trainer: Speech therapy plays key part in early intervention programs. Aliza provides games which are designed based on Augmentative and Alternative Communication (AAC) technique for speech development skills. There are three different games to learn alphabets, identify alphabets and objects and spell word. A Child has to identify the object or the alphabet shown on the screen and

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pronounce the word. The aim of this activity is to evaluate based on the number of words they managed to pronounce in a given time. Spell word game uses visual drag and drop interface to form the word by arranging the letters. Level increases up to five letter words based on the points. Speech to text model is integrated with the game to evaluate the user during the speech activities.

For each microphone input, the component will compute the MFCC and send the features to model for predictions. MFCC is a feature extraction method widely used in Speech recognition systems. Audio signal needs to be converted to speech signal by a finite number of measures to be understood by the computer. The numbers expressed in Mel Scale are more representative of human perception. Thus Mel Scale representation usually make computers understand speech more like human, less like machines [20]. Therefore, MFCC has been used for process of audio signal.

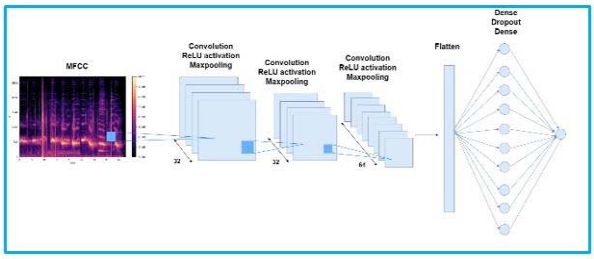


Fig. 4. CNN model for speech recognition

As show in the Fig. 4, first the Convolution layer performs filtering the input image which is the MFCC spectrogram to identify the features. After the convolution, Rectified Linear Unit (RELU) activation function is used to convert negative values to 0. This process adds non-linearity level to the network to train for classification and feature extraction. Max pooling followed by ReLU keeps the most important features and its approximate location. Convolution-ReLU-Max pooling process is repeated two times to get sixty-four feature maps. Finally, the feature maps are flattened into one to a long vector and fed into fully connected layer and final layer is given sigmoid activation function so that it will give binary (0 or 1) value to indicate confidence of the model for the MFCC image. Computed MFCC feature for the input voice of the user will be sent to this CNN model to classify the word. Classified word will be compared with existing list of words and finally, a report will be generated based on the progress of the child.

d) Attentiveness Tracker: Emotions are the major key of this component. Based on facial emotions the module tracks the user's attentiveness and responds in the game. Teaching according to the emotions of student has been proven very effective that applied behavior analysis recommends teaching according to autism children's emotions. Sensory Integration Therapy (SIT) entirely describes this teaching method [21].

Initially a maze game is introduced —for users for better improvement of their concentration and produced an auto generated report of their game play shown in Fig. 5, 6. When users are given the smart mirror to interact, a real-time emotion recognition model commences. Therefore, the teaching can fluctuate according to their emotions.

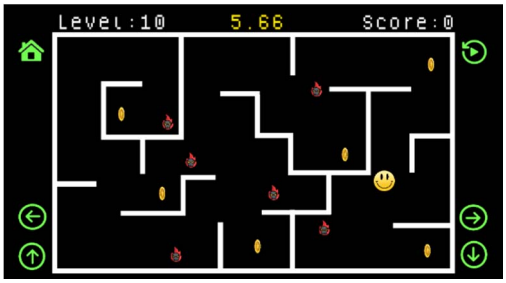


Fig. 5. Maze game level 10



Fig. 6. Maze game report

The proposed algorithm recognize emotions by,

1) Face detection using Viola-Jones detector
2) Captured image frames pre-processed and normalized
3) Apply 17 layers of Convolutional Neural Network (CNN) layers

According to the block diagram Fig. 7, using 'Haar Cascade' a fontal face is detected in each frame from the web camera input. Then the images are converted to 48 by 48 grayscale images. The pre-processed data filters through 17 sequential Convolutional neural network layers in order to ideally fit the model. In the convolutional layers, 'ELU' activation is used and for model compilation, 'Adam' optimizer is used. The overall trained model is flatten, compiled and optimized to demote the possibility of over fitting or under fitting. The saved model and model weights are then loaded to detect the actual face with emotions which is sent by the unity game. If negative emotions are detected the game prompts the user to swap to suggested activities of their predilection or if positive to continue with the same activity and track the concentration level of that game.

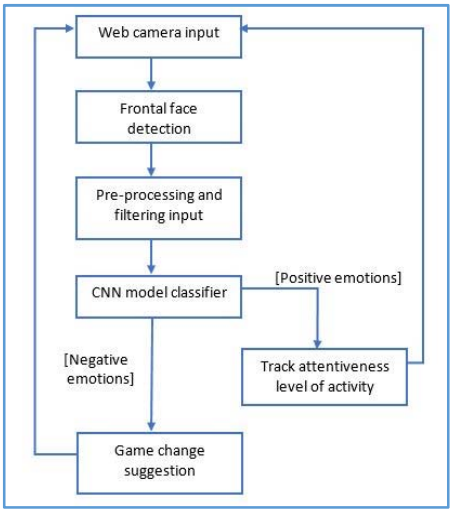


Fig. 7. Attentiveness tracker block diagram

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C. System Diagram

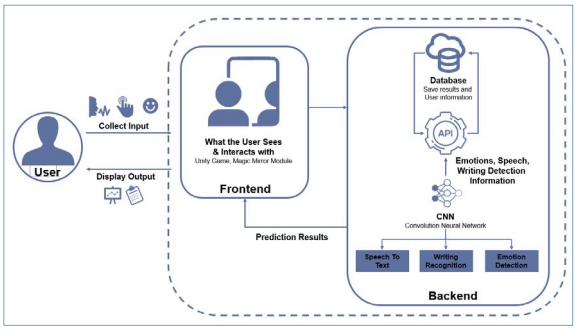


Fig. 8. Aliza System Diagram

Above mentioned pre-trained models are deployed using Flask as a web service. Relevant services are called within the unity application during the game play. Finally, processed model results and user details are saved in the database. The overall system diagram is as shown Fig. 8.

IV. Results and Discussion

A. Results and Output

Development of ASD childrens' basic education is the main goal of this research. Level of education provided should be in a high expected standard according to ASD therapies also these users acquire more guidance. Results from the ASD users are evaluated with the non ASD users to compare their improvement.

The first component Writing Mentor provides training to the user through the introduction game. After many levels of training, self-drawing of shapes is given to evaluate their progress of writing. The sketch area is taken as an image input and sent through a CNN model that is trained with ASD and non ASD children hand writing to predict the percentage of similarity.

TABLE I. Writing Menter test rates

Shapes	Success Rate (%)	No Of Test Run Images
HorizontalLine	70	10
PerpendicularLine	80	10
HalfCircle	70	10
Circle	70	10
Moon	60	10

The model testing returned an output accuracy of 80% which is an ideal test prediction of the writing mentor component. Given in Table I shows the rates of testing done with sample images of hand written shape in the model. Writing mentor predicts all types of shapes fed into the system and evaluates accordingly. Initially the writing mentor was trained with 1000 data samples using CNN model. Even though the accuracy was up to 84%. The number of layers were increased up to 7-layers, data samples increased by data augmentation which resulted a better and faster model.

2nd component of Aliza Math Tutor, Identifying numeric in the game accurately in a predicted rate of 85%. When testing done with autism student's hand written numbers the accuracy rate is as shown in Table II. 5 CNN layers trained with 1000 data correctly predicts all the occurrence of numeric. This model had a loss rate of 0.4 percent. Earlier the response time was 5s, in order to increase the responsiveness of math module the model was re-trained using the equivalent CNN model with additional of 3000 data samples.

TABLE II. Math Tutor test rates

Numbers	Success Rate (%)	No Of Test Run Images
0	80	15
1	90	15
2	80	15
3	86.66	15
4	73.33	15
5	80	15
6	80	15
7	86.66	15
8	66.66	15
9	73.33	15

Verbal Trainer is the 3rd component of Aliza. The speech-to-text model used for the evaluation in verbal trainer games is trained using Tensor flow speech command dataset which contains audios of 30 words, uttered by 2618 speakers that totally add up to 105,829 utterances. These are a set of one second audio files recorded at 16kHz in .wav format. The trained CNN model was tested against the test set provided in the dataset and scores accuracy of 90%. According to the Table III, the success rate obtained while beta testing in isolated area is comparatively less for some words. Since the frame size is 1 second and an overlap of 0.2 seconds is used, only the 0.8 seconds of the next audio input is taken while inferencing. Therefore, for longer words there is a relatively high probability for a part of the word to get trimmed which leads to decrease in accuracy for longer words compared to short words.

TABLE III. Verbal trainer test rates

Phrases	Success Rate (%)	No Of Test Run Audio
House	60	5
Cat	100	5
Cow	100	5
Lion	80	5
Cake	80	5

The attentiveness tracker final component of Aliza, consisting of 3-layer CNN model that predicts images in a rate of 90%. Model was capable to identify the emotion of the main face image. Sample images of emotions tested with created model showed test rates as given in Table IV. The model was built using Keras, with 1000 data examples. When beta testing this component the Classification model loss accuracy rate of 5% which indeed was a drastic loss impacting the main goal of the component. Examining the glitch the model was enhanced consisting 17-layer CNN model using 'ELU' activation instead of 'ReLU' to enforce the model to reach the accuracy of 89% with a loss of fair amount of 0.2%.Adam optimizer resulted as the best fit to model comparatively to SDG and RMSprop.

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TABLE IV. ATTENTIVENESS TRACKER TEST RATES

Emotions	Success Rate (%)	No Of Test Run Images
Happy	87.50	8
Suprise	75	8
Angry	75	8
Sad	87.50	8
Fear	62.50	8
Disgust	62.50	8
Neutral	100	8

V. CONCLUSION AND FUTURE WORKS

With the increase in autism children [22] there is a lack of resource to cater to the needs of all. Nevertheless, this research paper states how 'Aliza' smart mirror sets a new standard among the training and teaching methods. Therefore, this research holds training and teaching methods based on game activities which are helps to increase interest and focus of autism students and it accommodates basic education for age groups of three to ten aged children. This game device is built using all aspects of Applied Behavior Analysis (ABA) teaching technique. This includes an emotion tracking system that will help to track concentration level of the activities of Autism students. Unlike other systems available in the global market, 'Aliza' is at an affordable cost which can be consumed by single or multiple users and educate students based on their emotions for a better understanding of the programs given. Smart concept is currently adapted in many industries and enormously increasing the use in daily household. Therefore, same concept is adapted for autism education for a better change.

Due to the pandemic situation some limitations to the system were enforced. Users are expected to wear least amount of accessories since it misclassifies the expressions (E.g. scarf, mask). Covid-19 pandemic situation limited the data collection since schools and Autism Centre were closed and publicly available data was used to develop the system.

To maintain the product more sustainable and reliable some future enhancement works have been outlined to be achieved. The feature enhancements are mainly focused on supporting local languages such as Sinhala and Tamil, then to develop a facial recognition system for login. Also teaching moral values, health exercises, and social skills activities are added as value added services. The planned future work is not only to enhance the market value of the product nonetheless to be an innovative solution to all types of individuals in developing their basic education needs.

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